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Optical studies of the 1.945 eV vibronic band in diamond

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The 1.945 eV band is the dominant end product of annealing radiation damage in diamonds containing isolated substitutional nitrogen atoms. In this paper uniaxial stress experiments are described which show the band to be a transition between A_1 (ground) and E (excited) electronic states of a trigonal centre, the E state being coupled predominantly to A_1 modes of vibration. Annealing experiments suggest that the centre is either a substitutional nitrogen atom plus an interstitial nitrogen atom, or a substitutional nitrogen atom plus a vacancy. Small differences between the luminescence and absorption bands are shown to be consistent with the double minima vibrational potential of the nitrogen-vacancy model.

1. INTRODUCTION

In this paper we are concerned with the properties of the 1.945 eV vibronic band of diamond. This band is the dominant end product of annealing primary radiation damage in 'type 1b' diamonds (figure 1). Now type 1b diamonds are those diamonds whose major impurity is nitrogen which is present as isolated substitutional atoms (Dyer, Raal, du Preez & Loubser 1965). Therefore the 1.945 eV centre is likely to be very simple – a radiation damage centre trapped at a substitutional nitrogen atom – and so it is well worth a detailed study.

Properties of the 1.945 eV band have been given in three earlier studies. The first, by du Preez (1965), gave the necessary conditions for observing the band, namely radiation damage of a diamond containing isolated nitrogen atoms, followed by annealing at temperatures over 900 K. In the second paper, Clark & Norris (1971) showed that polarized luminescence measurements limited the symmetry of the centre to trigonal or monoclinic I or rhombic I. The third paper reported the temperature dependence of the 1.945 eV absorption band (Davies 1974) and will be briefly referred to in §§ 6 and 7.

This paper reports further detailed studies of the centre. In § 3 we use annealing studies to show that the centre is probably either a vacancy-substitutional nitrogen pair, or an interstitial nitrogen-substitutional nitrogen pair. In §§ 4, 5 uniaxial stress experiments are described which conclusively establish the symmetry as trigonal, the zero phonon line being between A_1 and E states. The vibronic coupling is found to be predominantly to A_1 modes. In § 6 small deviations of mirror symmetry in the

absorption and luminescence spectra are noted. Finally, all these data are shown to be consistent with the properties of a simple model of the vacancy-substitutional nitrogen pair.

2. SAMPLES AND EQUIPMENT

Samples were irradiated at room temperature with approximately $5 \times 10^{21} \text{ m}^{-2}$ 2 MeV electrons from the University of Reading van der Graaff accelerator, and then annealed, typically at 1100 K for 2 h. In type 1b diamonds the 1.945 eV centres could then be observed in absorption (figure 1c). In type 1a diamonds (those with the nitrogen impurity predominantly present in aggregates of substitutional nitrogen atoms) the weak absorption spectrum was more readily measured indirectly by the luminescence excitation technique. Photoluminescence was generated in the types 1a and 1b samples by irradiation with the yellow lines from a high pressure mercury lamp.

Uniaxial stresses were applied via simple push-rods driven by oil pressure. The samples were in metallic contact with either liquid nitrogen, for the data of § 4, or liquid helium, for § 5. Spex 1701 and 1702 monochromators were used with photomultiplier detectors (E.M.I. 9558 QB). The signal was d.c. amplified for presentation on a chart recorder or output on a punched tape for computer processing.

3. ANNEALING EXPERIMENTS

Irradiated type 1b diamonds have been carefully annealed in the hope that the destruction of one radiation induced centre could be matched with the growth of the 1.945 eV band, thereby identifying its constituents.

Before irradiation the visible spectrum of type 1b diamond is characterized by a continuous absorption (figure 1a) whose strength is proportional to the concentration of isolated nitrogen atoms. In the type 1b specimens used here the nitrogen concentration, measured from the absorption coefficient at 2.64 eV before irradiation, was *ca.* $4 \times 10^{24} \text{ m}^{-3}$ (Sobolev *et al.* 1969; Chrenko, Strong & Tuft 1971). Room temperature irradiation with 2 MeV electrons produces easily observable zero phonon lines at 1.673, 1.860, 1.979, 2.085, 2.367, 2.462, 2.535 and 3.150 eV (figure 1b). Annealing for a few minutes at 800 K destroys all but the 1.673 and 3.150 eV bands with no measurable 1.945 eV absorption being produced. Annealing at temperatures above 900 K reduces the 1.673 and 3.150 eV lines, the lines disappearing at similar rates to each other in both isochronal and isothermal annealing experiments, as, for example, in figure 2. Simultaneously the 1.945 eV line is produced (figure 2), strongly suggesting that either the 1.673 or the 3.150 eV centre is involved in the 1.945 eV centre: correlation tests have not convincingly determined which. A tentative model for the 1.945 eV centre is thus either a 1.673 eV or a 3.150 eV centre trapped near a substitutional nitrogen (since the 1.945 eV centre is only readily observable in type 1b diamond). It is unlikely that *both* the 1.673 and the 3.150 eV centres are

required in view of the trigonal symmetry of the 1.945 eV centre (§ 4, below). The concentration of 1.945 eV centres obtained after fully annealing the type 1b diamonds used here was *ca.* $3 \times 10^{23} \text{ m}^{-3}$ for the usual dose of *ca.* $5 \times 10^{21} \text{ m}^{-2}$ 2 MeV electrons. This estimate results from detailed balance calculations (Fowler 1968) assuming a radiative lifetime of 10^{-8} s for the transition, which is allowed electric

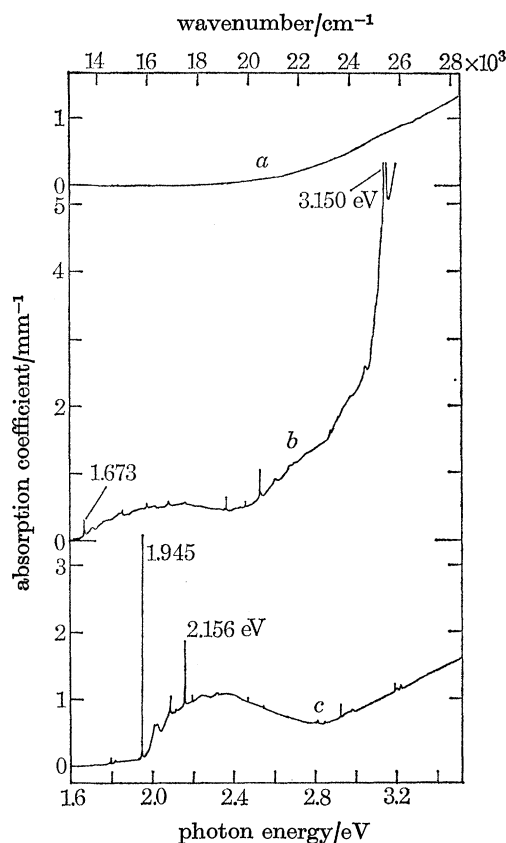


FIGURE 1. Visible absorption spectra at liquid nitrogen temperature of three similar blocks of type 1b diamond cut from same stone. (a) Unirradiated. (b) Irradiated at room temperature with *ca.* 10^{22} m^{-2} 2 MeV electrons. The extremely large ratio of 3.150 to 1.673 eV absorption is characteristic of type 1b diamond. (c) Irradiated at room temperature with *ca.* $5 \times 10^{21} \text{ m}^{-2}$ 2 MeV electrons and then annealed as shown in figure 2. The relatively intense 2.156 eV line is a result of the long annealing at low temperature (cf. figure 6). All spectra were taken using unfiltered light from a 100 W quartz-iodide lamp incident on specimens.

dipole (§ 4). The estimate is consistent with the total concentration of primary damage centres of about 10^{24} m^{-3} expected from Collins' work (1974) on the damage rate in diamond.

More detailed annealing experiments are not justified in view of the uncertainties caused by charge transfer processes. Photons absorbed by the 3.150 eV centres

liberate free charges (Farrer & Vermeulen 1972) some of which are captured at 1.673 eV centres (Dyer & du Preez 1965; Davies 1975). This weakens the 3.150 eV line and strengthens the one at 1.673 eV. Blue light, $2.88 < h\nu < 3.15$ eV, gives the reverse effect through absorption into a series of transitions at the 1.673 eV centre known as the GR2, GR3, ..., GR8 lines (Clark, Ditchburn & Dyer 1956; Walker, Vermeulen & Clark 1974). Even when the 1.673 and 3.150 eV centres have been annealed out, the 1.945 eV line can be decreased 30 % in strength by 365 nm radiation from a high pressure mercury lamp, possibly because light of $h\nu > 1.7$ eV

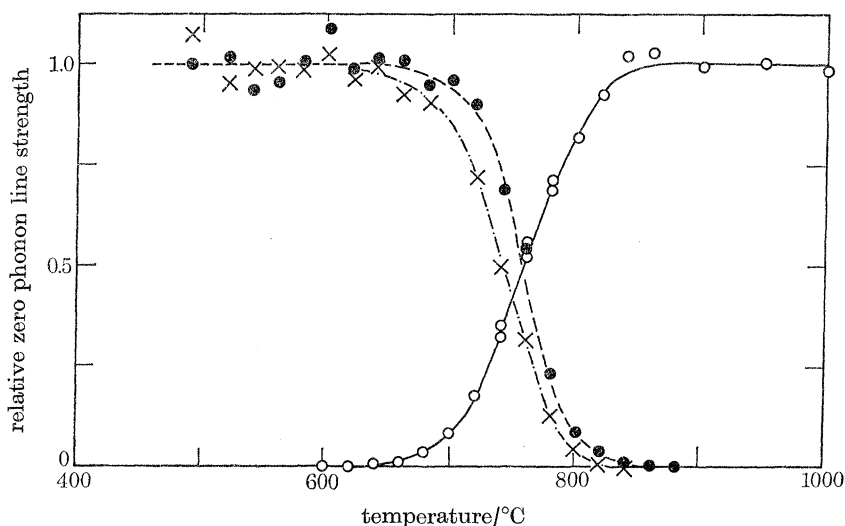


FIGURE 2. Result of isochronal annealing (30 min at each temperature) of an electron irradiated type Ib diamond. Zero phonon line strengths are shown for 1.673 eV (●), 3.150 eV (×) and 1.945 eV (○) lines, scaled to unit maximum strengths. Final state of diamond shown in figure 1c.

ionizes the isolated nitrogen atoms (Farrer 1969). The data in figure 2 were taken under reproducible conditions of illumination but, even so, their precise meaning is not clear. We simply conclude that the 1.945 eV centre appears to be a substitutional nitrogen atom plus either a 1.673 eV centre or a 3.150 eV centre. Annealing experiments in type 1b diamonds are non-trivial.

The 1.673 eV centre is a vacancy (see, for example, Mitchell 1967) and the 3.150 eV centre is probably an interstitial nitrogen atom (Davies & Lightowlers 1970). Either centre trapped in the nearest position to a substitutional nitrogen atom will produce a trigonal centre. This is confirmed in the next section.

4. UNIAXIAL STRESS EFFECTS ON THE ABSORPTION ZERO PHONON LINE

Uniaxial stresses of up to 2 GPa have been applied along the $\langle 001 \rangle$, $\langle 111 \rangle$ and $\langle 110 \rangle$ axes with the crystals at liquid nitrogen temperature. The effects on the zero phonon absorption line are completely straightforward and are illustrated by figures 3 and 4. Tables of the splitting patterns show that our results are consistent only with an electric dipole transition between isolated A_1 and E states at a trigonal centre, the ground state being A_1 (Kaplyanskii 1964*a, b*; Hughes & Runciman 1967). The theory of the stress splitting of this type of transition has been given by, for

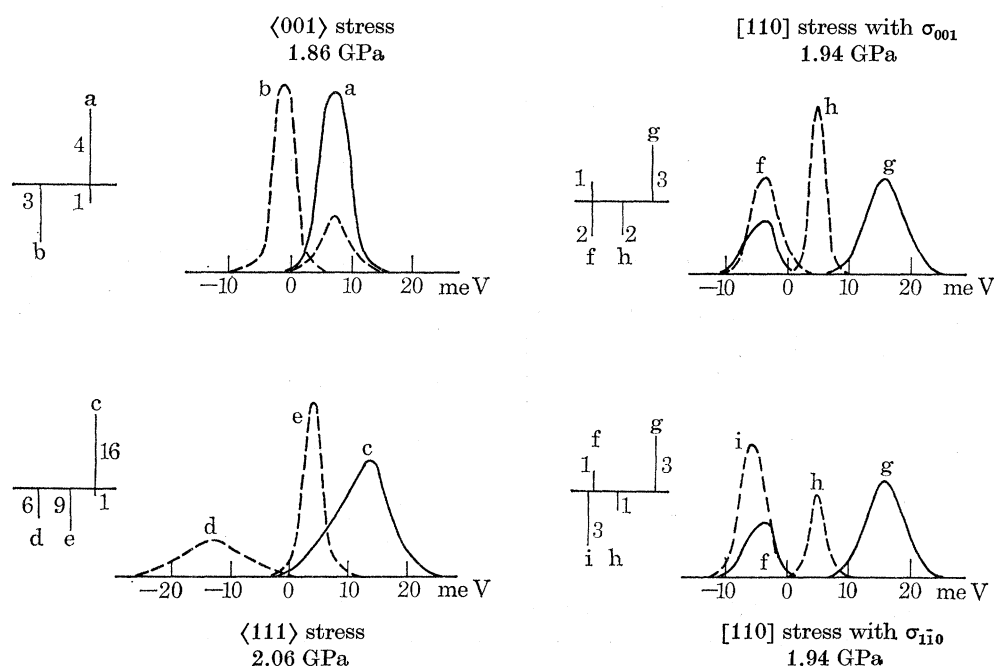


FIGURE 3. Effect of uniaxial stress on the 1.945 eV zero phonon absorption line measured at liquid nitrogen temperature. Solid lines are for π -polarization, broken lines for σ -polarization. The line spectra show the intensities of the stress-split components of an A_1 to E transition at a trigonal centre.

example, Hughes & Runciman (1967) and Crowther & Dean (1967). The theory is outlined here to fix the notation.

We take as the basic trigonal centre one that is oriented along $[111]$, defined as the Z axis. The X and Y axes are defined as $[11\bar{2}]$ and $[1\bar{1}0]$, respectively. Y is then the normal to a reflexion plane of the centre. In terms of the stresses s_{ij} defined with respect to the crystal axes x, y, z the stress perturbation is

$$\begin{aligned}
 V = & A_1(s_{xx} + s_{yy} + s_{zz}) + A'_1(s_{yz} + s_{zx} + s_{xy}) \\
 & + E_X(s_{xx} + s_{yy} + 2s_{zz}) + E_Y\sqrt{3}(s_{xx} - s_{yy}) \\
 & + E'_X(s_{yz} + s_{zx} - 2s_{xy}) + E'_Y\sqrt{3}(s_{yz} - s_{zx}).
 \end{aligned}$$

The operators $A_1, E_X, \dots$ transform as A_1, E_X , etc. Stress coupling coefficients are defined in terms of the double barred matrix elements of the operators $A_1, A'_1, \dots$ between the $|A\rangle, |EX\rangle$ and $|EY\rangle$ states of the centre by

$$A_1 = [E||A_1||E] - [A||A_1||A], \quad 2A_2 = [E||A'_1||E] - [A||A'_1||A],$$
$$B = [E||E||E]/\sqrt{(2)}, \quad C = [E||E'||E]/\sqrt{(2)}.$$

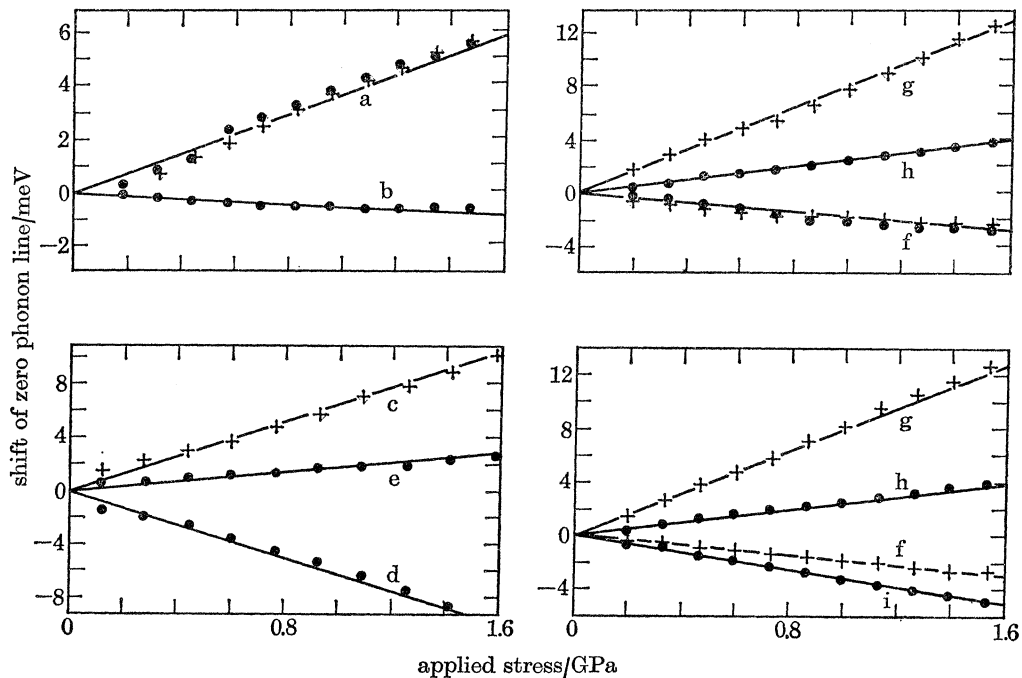


FIGURE 4. Energies of the stress-split components as functions of stress. Data shown by + for π - and $\bullet$ for σ -polarizations. Lines are least squares fits of straight lines.

TABLE 1. EFFECTS OF STRESS ON ZERO PHONON ABSORPTION LINE

stress axis	line	final state	experiment	$10^{12} \times$ shift rate (eV Pa ⁻¹)	theory
001	a	$ EX\rangle$	3.80	$A_1 + 2B$	3.54
	b	$ EY\rangle$	-0.45	$A_1 - 2B$	-0.61
111	c	$ EX\rangle$	5.89	$A_1 - \frac{2}{3}A_2 + \frac{4}{3}C$	6.29
	d	$ EX\rangle, EY\rangle$	-6.30	$A_1 + 2A_2$	-6.23
	e	$ EY\rangle$	1.61	$A_1 - \frac{2}{3}A_2 - \frac{4}{3}C$	1.78
110	f	$ EX\rangle$	-1.61	$A_1 + A_2 + C - B$	-1.73
	g	$ EY\rangle$	8.19	$A_1 - A_2 + C + B$	8.04
	h	$ EX\rangle$	2.68	$A_1 - A_2 - C - B$	2.59
	i	$ EY\rangle$	-3.17	$A_1 + A_2 - C + B$	-3.03

Stress coefficients: $A_1 = 1.47 \times 10^{-12}$ eV Pa⁻¹, $A_2 = -3.85 \times 10^{-12}$ eV Pa⁻¹, $B = 1.04 \times 10^{-12}$ eV Pa⁻¹, $C = 1.69 \times 10^{-12}$ eV Pa⁻¹.

The stress effects are summarized in table 1. The expressions are similar to those of Hughes & Runciman (1967) except that the signs of B and C are reversed. This difference is a result of the $[1\bar{1}0]$ direction in diamond not lying in a reflexion plane of the standard trigonal centre, in contrast, for example, to the situation in the sodium chloride structure.

The nine observed shift rates are closely described (table 1) by using the values of A_1 , A_2 , B and C listed in the table.

5. UNIAXIAL STRESS MEASUREMENTS ON THE LUMINESCENCE BAND

The one phonon sideband of the luminescence band has been studied under $\langle 001 \rangle$ stress to see if the E excited state undergoes a Jahn–Teller distortion through coupling to E modes of vibration. The method is that of Fetterman & Fitchen (1968).

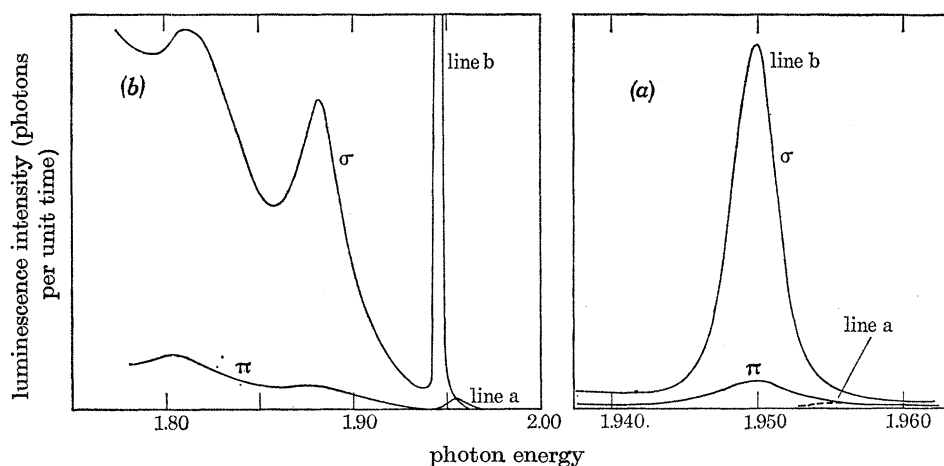


FIGURE 5. 1.945 eV luminescence band under an $\langle 001 \rangle$ stress of 14×10^8 Pa at 17 K. (a) Experimental data for zero phonon line. (b) One phonon and part of two phonon sidebands after correction for random strain depolarization present in (a).

Under $\langle 001 \rangle$ stress the lowest vibronic state derived from the E electronic state is that which transforms as $|EY\rangle$ in the unperturbed system (table 1). Electric dipole transitions from $|EY\rangle$ to a vibronic state transforming as A_1 can only occur in the σ polarization—for example, line b of figure 3.† An E mode of vibration in the A_1 electronic state gives a total vibronic state transforming as E. Transitions to this E state from $|EY\rangle$ are group-theoretically allowed with the electric vector along X , Y or Z . Assuming that the matrix elements for the transitions may be factorized into a purely electronic part which contains the electric vector and a purely vibrational part, the transition probability under Z polarized light is zero. Transitions

† Throughout, π means electric vector parallel to the stress axis, and σ perpendicular. σ_{hkl} means electric vector along the $[hkl]$ axis.

from $|EY\rangle$ to an E vibronic state then appear equally in π - and σ -polarizations under the $\langle 001\rangle$ stress.

This ideal situation is modified by random stresses in the crystal which mix the $|EX\rangle$ and $|EY\rangle$ eigenstates of the $\langle 001\rangle$ stress perturbation. To estimate the effect on the spectra we represent the random stresses by a single stress r which is directed along one of the major crystallographic axes. The magnitude of r is sufficient to split the E state of the standard $[111]$ oriented centre by an amount equal to the observed full width at half height, Γ , of the zero phonon line ($\Gamma = 2.5$ meV). For example, when r is directed along $[\bar{1}11]$ its magnitude is

$$r = 3\Gamma/80 = 5.55 \times 10^8 \text{ Pa.}$$

For the experimental results of figure 5 an external stress of $s_{zz} = 14 \times 10^8$ Pa was applied along $[001]$. The combined stresses s_{zz} and r along $[\bar{1}11]$ split the E state. The lower split-off level is found, from the secular matrix, to be not purely $|EY\rangle$ but

$$|EY'\rangle = 0.98 |EY\rangle + 0.22 |EX\rangle.$$

Transitions from $|EX\rangle$ to a ground state transforming as A_1 are strongly allowed in the π -polarization (i.e. electric vector along $[001]$), as for line a of figure 3. The intensities I_π and I_σ of the $|EY'\rangle$ to A_1 transitions observed in the π - and σ -polarizations are therefore non-zero: calculation gives

$$I_\pi/I_\sigma = 0.053. \quad (1)$$

Similar depolarization is predicted when r is orientated along axes other than the $[\bar{1}11]$ case discussed here.

Experimentally, the zero phonon line has (figure 5a)

$$I_\pi/I_\sigma = 0.065.$$

This is sufficiently close to the estimate in equation (1) that we may dismiss the depolarization as simply being a result of random stresses.

The one phonon sideband is predominantly σ -polarized (figure 5b). Correction for the random stress depolarization gives $I_\pi/I_\sigma \approx 0.1$. Thus about 10 % of the one phonon sideband is due to E modes and so 90 % is caused by A_1 modes.

6. COMPARISON OF THE ABSORPTION AND LUMINESCENCE BANDS

The electrons of the 1.945 eV centre are predominantly coupled to A_1 phonons (§ 5). The temperature dependence of the zero phonon line is similar to other lines in diamond whose properties can be described by assuming the coupling is linear in the vibrational displacement, with only *small* quadratic terms (Davies 1974). The 1.945 eV absorption and luminescence bands would therefore be expected to be 'mirror images' in the zero phonon line. That is, the luminescence intensity $I(\nu_0 - \nu)$

in photons per second at a frequency ν below the zero phonon frequency ν_0 is expected to be related to the absorption $\mu(\nu_0 + \nu)$ by

$$I(\nu_0 - \nu) = \mu(\nu_0 + \nu) (\nu_0 - \nu)^3 / (\nu_0 + \nu). \quad (2)$$

Figure 6 compares the measured spectrum with $I(\nu_0 - \nu)$ calculated from $\mu(\nu_0 + \nu)$.

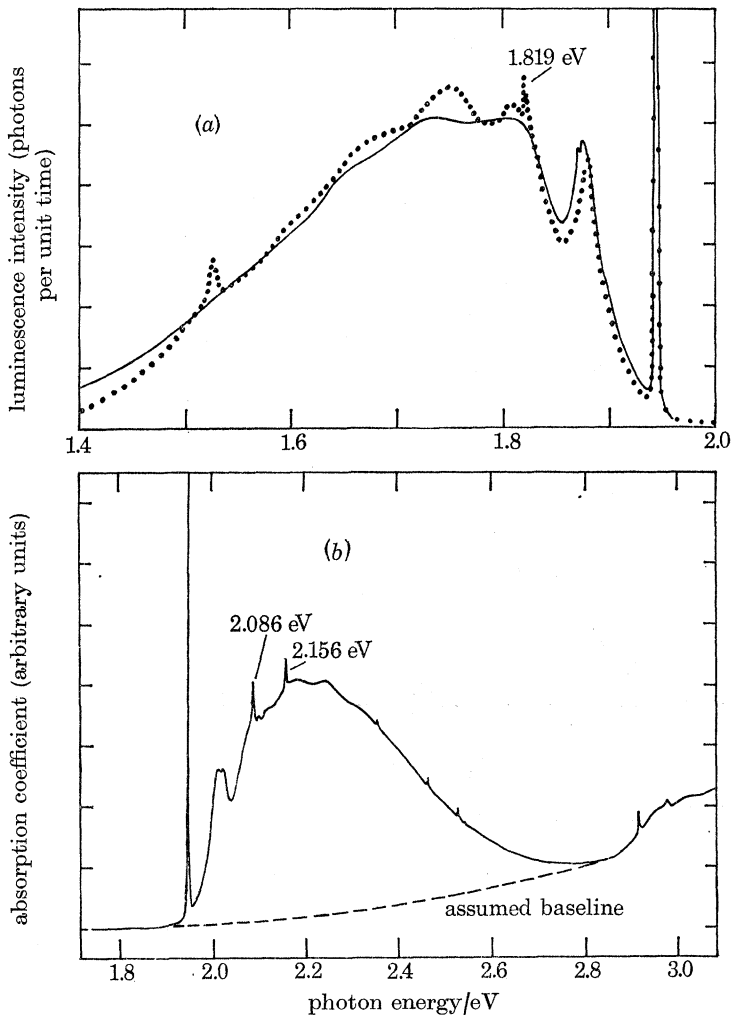


FIGURE 6. Luminescence spectrum measured at liquid-nitrogen temperature ($\bullet$) compared with the luminescence spectrum calculated from the absorption spectrum using equation (2). The absorption spectrum is shown in (b) together with the assumed baseline. This diamond had been electron irradiated and annealed for 2 h at 1100 K; cf. figure 1c.

There is general agreement as to the overall shape when the independent transitions at 1.819, 2.086 and 2.156 eV are ignored. However, the 2 and 3 phonon sidebands are less well resolved in the absorption band than in the emission band. Secondly the one phonon sideband is a single peak in the luminescence spectrum but a

doublet in absorption. This result has been checked in natural and synthetic type 1b diamonds and in a type 1a diamond using luminescence excitation to measure the very weak absorption.

In the next section we develop a model that accounts for these differences.

7. VIBRONIC PROPERTIES OF A SUBSTITUTIONAL NITROGEN, VACANCY PAIR

Two equally likely models are consistent with the annealing and stress work: a substitutional nitrogen atom, N, plus either a vacancy, V, or an interstitial nitrogen atom in the nearest possible positions (§§ 3, 4). In the N-V model the

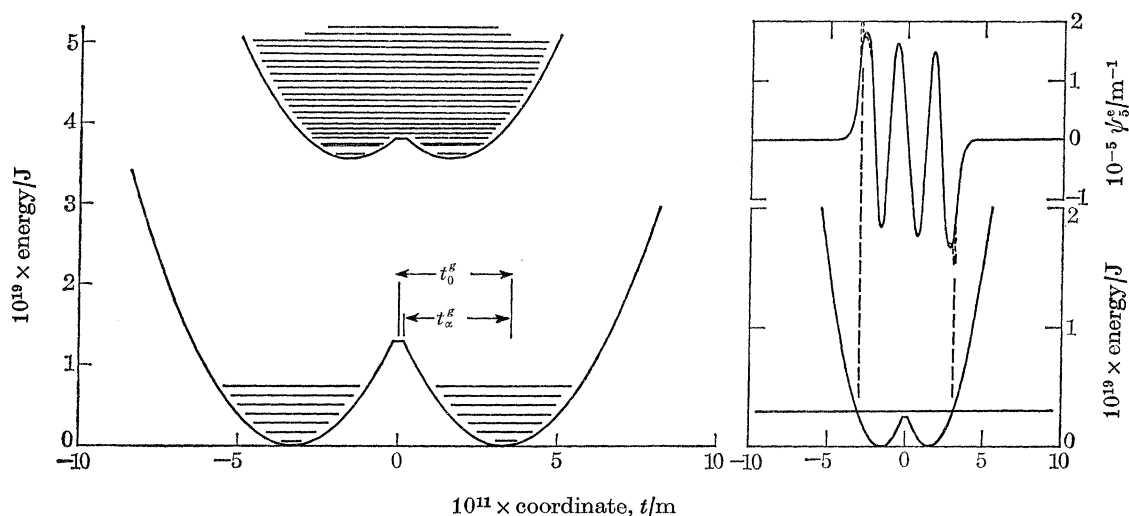


FIGURE 7. (a) Calculated adiabatic potentials and energy eigenvalues for the nitrogen-vacancy model. (b) Divergence of W.K.B. approximation wavefunctions is shown by broken lines in upper diagram for the $n = 5$ vibrational level of the electronic excited state. Small modifications give wavefunction shown by solid line with energy (from equation (4)) within 0.1% of the W.K.B. value (equation (3)).

N atom could tunnel into the vacancy, giving an equivalent V-N centre (see figure 7). The tunnelling motion provides a mechanism for splitting the one phonon sideband of the absorption spectrum into the observed doublet. In this section we study this model in some detail to see whether it is fully consistent with all the features of the observed spectra.

We assume the vibrational properties may be parameterized by a single mode in which the N atom is moving along the N-V axis. This is a totally symmetric mode, as required by § 5. Neglect of other types of vibration will overemphasize any peculiar vibronic properties of the N-V model. The total wavefunction for the centre

is written as a Born–Oppenheimer product, the vibrational potential being (see figure 7)

$$\begin{aligned} V^i(t) &= k(t + t_0^i)^2, \quad t < -t_0^i + t_\alpha^i \\ &= k(t_\alpha^i)^2, \quad -t_0^i - t_\alpha^i \leq t \leq t_0^i - t_\alpha^i \\ &= k(t - t_0^i)^2, \quad t > t_0^i - t_\alpha^i. \end{aligned}$$

The origin of t lies at the mid-point of two lattice sites, the N atom being in equilibrium (classically) at $t = \pm t_0^i$ in the i th electronic state. The curvature of the parabolae is determined by the vibrational quantum $\hbar\omega$ measured from the emission spectrum and the reduced mass ξ of the oscillator by

$$k = \frac{1}{2}\xi\omega^2, \quad \xi \gtrsim 3m_C m_N / (m_N + 3m_C).$$

Here m_C is the atomic mass of carbon, m_N of nitrogen.

In the excited electronic state ($i = e$) the first excited vibronic state is split by 10 meV while the zero phonon level is split by less than the zero phonon linewidth (*ca.* 2.5 meV), figure 6*b*. For our oversimplified model these splittings have been rounded off respectively to $\delta_1 = 0.1\hbar\omega$, $\delta_0 = 0.01\hbar\omega$. The precise value of δ_0 is not critical. The mass ξ has been equated to m_N and $\hbar\omega$ rounded off to 70 meV. Dennison & Uhlenbeck (1932) have given exact expressions relating t_0^i and t_α^i to δ_0 , δ_1 , ξ and $\hbar\omega$ (their equations 11) from which

$$t_0^e = 0.0156 \text{ nm}, \quad t_\alpha^e = 0.0137 \text{ nm}.$$

Since $V^e(t)$ has an even parity in t the vibrational wavefunctions ψ_n^e are even or odd in t . As transitions are observed to both split levels of the first excited state the vibrational wavefunction for the zero phonon level in the electronic ground state cannot just have even parity as it would for a single parabolic potential. $V^g(t)$ for the ground state must also have two minima, say of separation $2t_0^g$. At low temperature, $\exp(-3.65)$ of the total transition probability of the 1.945 eV band lies in the zero phonon line (Davies 1974). The relative displacements t_0^e and t_0^g are therefore, as in simple vibronic theory (see, for example, Hughes 1966, § 4.3)

$$|t_0^e - t_0^g| = (3.65\hbar\omega/k)^{\frac{1}{2}} = 0.0177 \text{ nm}.$$

$V^e(t)$ and $V^g(t)$ are now fully determined (figure 7). We note $2t_0^g = 0.0666 \text{ nm}$ is reasonable in that it does not exceed the atomic spacing of diamond, 0.154 nm.

Since the potential minima in $V^g(t)$ are well separated, each energy level of $V^g(t)$ is doubly degenerate and the vibrational wavefunctions are the sum and difference of harmonic oscillator functions centred on each minimum.

In the W.K.B. approximation the energy levels E_n for $V^e(t)$ are those for which the action is quantized (Bohm 1960)

$$\int_{t_1}^{t_2} \{2\xi(E_n - V^e(t))\}^{\frac{1}{2}} dt = (n + \frac{1}{2})\hbar\pi, \quad n \text{ integral.} \quad (3)$$

The expression applies to those states above the plateau near $t = 0$, i.e. when $E_n > k(t_\alpha^e)^2$. The integral is confined to values of t between t_1 and t_2 for which

$E_n \geq V^e(t)$. The energies are on figure 7. The W.K.B. vibrational wavefunctions are, for $t < 0$,

$$\psi_n^e(t) = A_n \exp \left\{ - \int_t^{t_1} p_1 dt \right\} / \hbar p_1^{\frac{1}{2}}, \quad p_1^2 = 2\xi(V^e(t) - E_n),$$

when $E_n < V^e(t)$, and

$$\psi_n^e(t) = 2A_n \cos \left\{ \int_{t_1}^t p_2 dt / \hbar - \frac{1}{4}\pi \right\}, \quad p_2^2 = 2\xi(E_n - V^e(t)),$$

when $E_n > V^e(t)$. Since $V^e(t)$ has even parity in t , the wavefunctions for $t > 0$ are $\psi_n^e(t) = (-)^n \psi_n^e(-t)$. The divergence in $\psi_n^e(t)$ at t_1 and t_2 has been smoothed out (figure 7), the criterion for correct smoothing being that

$$\int_{-\infty}^{\infty} \overline{\psi_n^e} \left[-\frac{\hbar^2}{2\xi} \frac{d^2}{dt^2} + V^e(t) \right] \psi_n^e dt \quad (4)$$

was within $\pm 0.3\%$ of E_n . The constants A_n in the wavefunctions ψ_n^e were chosen to normalize the smoothed ψ_n^e .

The probability of a transition between the m th vibrational level of the ground electronic state and the n th vibrational level of the excited electronic state was calculated as the square of the overlap integral of the two vibrational wavefunctions

$$T_{mn} = \left| \int_{-\infty}^{\infty} \overline{\psi_m^g} \psi_n^e dt \right|^2.$$

The low temperature spectra are given in figure 8. As a check on the spectra we note that ψ_n^e can be expanded as an infinite sum of harmonic oscillator wavefunctions $\chi_l(t)$ of frequency ω . From simple vibronic theory, transitions to the $\chi_l(t)$ from the essentially degenerate lowest vibrational levels ψ_0^g and ψ_1^g of the ground electronic state obey

$$\sum_{l=0}^{\infty} \left| \int_{-\infty}^{\infty} \overline{\psi_0^g} \chi_l dt \right|^2 + \sum_{l=0}^{\infty} \left| \int_{-\infty}^{\infty} \overline{\psi_1^g} \chi_l dt \right|^2 = 2.$$

(For example, see, Maradudin 1966, equation (13.13a).) It follows from the principle of spectroscopic stability that

$$\sum_{n=0}^{\infty} T_{on} = 2.$$

The W.K.B. calculation gives 2.02 for this sum, showing it to be a good enough approximation here.

The calculated absorption spectrum (figure 8) has more transitions per unit energy than the luminescence spectrum and will appear less structured in the 2 and 3 phonon region. The one phonon absorption sideband is a doublet, and the calculated absorption and luminescence spectra have similar half-widths. These results are as required by experiment, § 6.

Unfortunately the details of this vibronic model cannot be verified by cluster model calculations using the measured stress coupling coefficients. The calculations,

which for brevity will not be described, predict a large Jahn–Teller coupling, in contrast with experiment, § 5. Evidently the cluster model prerequisites (see, for example, Hughes 1970), of highly localized optical electrons at a centre which does not strongly perturb the crystal, are invalid at this covalently bonded centre.

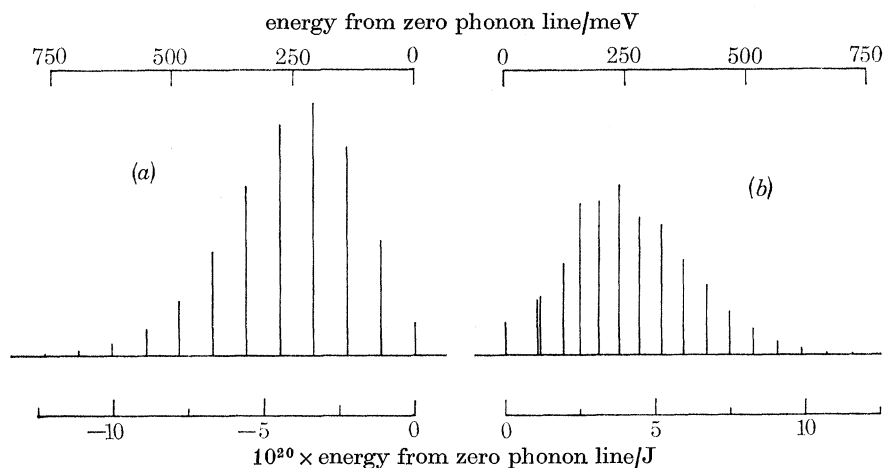


FIGURE 8. Calculated transition probabilities at 0 K for (a) luminescence and (b) absorption spectra of nitrogen-vacancy model. The spectra are normalized to the same intensity. The zero phonon lines are coincident in energy. The first 30 transitions of the absorption spectrum have been calculated.

8. SUMMARY

Uniaxial stress measurements on the 1.945 eV absorption band have shown it is caused by an A_1 to E transition at a trigonal centre (§ 4), in agreement with the less precise polarized luminescence results of Clark & Norris (1971). Small differences in the absorption and luminescence spectra (§ 6) have been shown to be consistent with a substitutional nitrogen, vacancy pair (§ 7) and this, in turn, is consistent with annealing studies of the centre (§ 3).

It is difficult to see how this model can be tested further by optical techniques. Electron paramagnetic resonance may be helpful since the charge state of the 1.945 eV centre can apparently be changed by ultraviolet radiation (§ 3). At least one charge state would be expected to be e.p.r. active. Whether the signal can be separated from that of the isolated nitrogen atoms—which also have $\langle 111 \rangle$ axial symmetry—remains to be seen.

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